Multi-messenger Astronomy: Optical Follow-up of a Short Gamma-ray Burst

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Abstract

We used the 0.4318 m ground-based telescope at the University of Texas Rio Grande Valley's Dr. Cristina V. Torres Memorial Astronomical Observatory (CTMO) located in Resaca de la Palma State Park in Brownsville, Texas. The CTMO telescope is an f/6.8 CDK17 astrograph manufactured by PlaneWave. We used an FLI ProLine CCD with no filter. On the night of 2024 June 16, CTMO pursued a short gamma-ray burst GRB240615a via optical observational methods. All observations were 60 second exposures set at 10°C and binned in 1 x 1 mode. Information regarding the transient, such as its coordinates, was provided through the NASA GCN alert system. A GCN circular for GRB240615a was released when the Fermi Gamma-ray Burst Monitor (GBM) was triggered on 2024 June 15 at 17:51:45 UT. We followed-up the alert provided by Swift-BAT GUANO. Data reduction and photometry was performed by Python scripts and catalog cross-matching by ds9. We did not observe GRB240615 in the optical spectrum using the CTMO telescope.

Introduction



Fig. 3: The Fermi Gamma-ray Burst Monitor (GBM)

In 1916, Albert Einstein predicted the existence of gravitational waves by using equations from his Theory of General Relativity. Gravitational waves are ripples in the fabric of spacetime, caused by events with mass and energy. High-energy events such as supernovae, binary black hole systems, and neutron star mergers (kilonovae), can produce gravitational waves. Figure 1 depicts the behavior of the gravitational waves produced from a neutron star merger during the stages of its inspiral and eventual collision.

Additionally, kilonovae emit detectable signatures in another form called a gamma-ray burst (GRB). As seen in Fig. 2, the gamma-ray burst is an attribute of the kilonovae's collimation jet. The collimation jet prompts an afterglow, emitting electromagnetic counterparts that are detectable by ground-based facilities and space observatories, depending on the spectrum. An example of a space detector is the Fermi GBM (Fig. 3). The Fermi GBM can localize and derive spectral information from a signal as a resource in space, while CTMO's mission is to execute optical ground-based follow-up observations.

Methods

- **Observations:** To use the alerts from NASA GCN to locate and observe GRB240615a in the optical spectrum. The hardware apparatus consists of mainly the PlaneWave astrograph (Figure 4) and the FLI ProLine CCD. Software includes PlaneWave for telescope controls, Maxim DL for CCD operations, and Stellarium to locate objects in the night sky.
- **Analysis:** The CTMO pipeline are Python scripts that calibrate and perform photometry using the science images. A GRB can be detected by catalogue cross-matching a list of observed magnitudes and a known survey. Ideally, if the GRB is present, then a light curve of its brightness over time can be produced.

Objectives

- **The Big Idea:**
  - Observing the afterglow of a kilonova can provide an insight on how heavy/rare Earth metals are created in the universe. The coalescence of compact, neutron-abundant objects triggers what is known as the r-process, or rapid neutron capture process; which, results in lanthanides.
  - Apart from its investigatory role in ascertaining the physical properties of kilonovae, observing GRBs can help determine how frequently they occur.
  - A light-curve from the GRB afterglow can be produced using filters, as seen in Fig. 8.
  - **CTMO's purpose:**
    - The objective of this project is to observe a GRB in the optical spectrum and analyze the obtained data to produce i) a catalog of observed magnitudes for cross-matching, including the limiting magnitude and ii) a magnitude-color diagram.

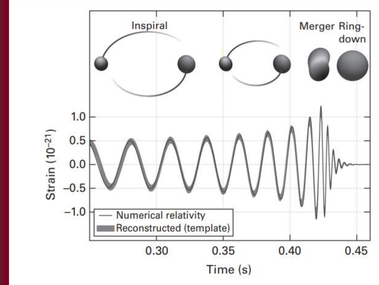


Fig. 1: The observed change in strain and frequency of the gravitational waves detected throughout the stages of a neutron star merger [7].

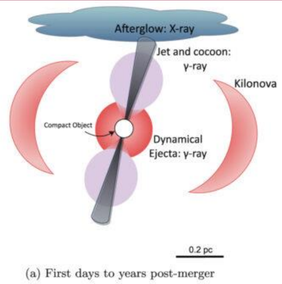


Fig. 2: Kilonova signatures/remnants. The afterglow produces electromagnetic counterparts [3].

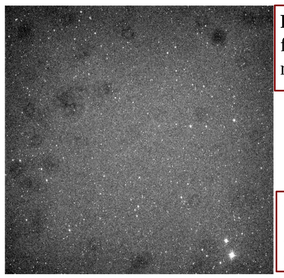


Fig. 4: (*Left*) Science image from the GRB240615 observing run, uncalibrated.

Fig. 5: (*Right*) Science image from the GRB240615 observing run, calibrated.



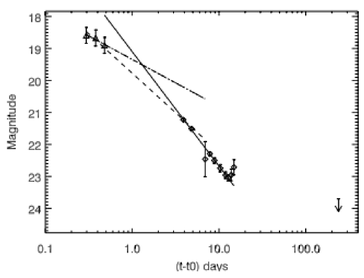


Fig. 8: Light curve of the GRB 030725 afterglow in the Rc-band [5].

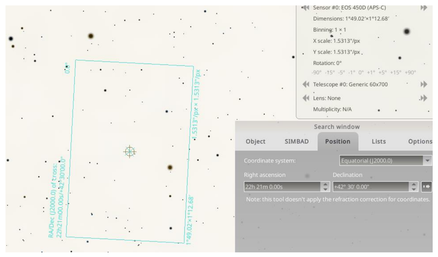


Fig. 9: A visual representation of where GRB240615 would be located in the night sky using Stellarium. GRB240615 was reported to have an RA of 22h 21m and a Dec of 42° 30'.

References

[1]A. von Kienlin, "Fermi GBM," *www.mpe.mpg.de*, 2024. https://www.mpe.mpg.de/617954/Fermi-GBM (accessed Aug. 01, 2024).
[2]A. Volnova *et al.*, "GRB 240615A: SAI CMO J- band observations," *GRB Coordinates Network*, vol. 36777, p. 1, Jun. 2024, Accessed: Aug. 01, 2024. [Online]. Available: https://ui.adsabs.harvard.edu/abs/2024GCN.36777....1V/abstract
[3]D. Finley, "Radio Observations Confirm Superfast Jet of Material From Neutron Star Merger," *National Radio Astronomy Observatory*, Sep. 15, 2018. https://public.nrao.edu/news/superfast-jet-neutron-star-merger/ (accessed Aug. 01, 2024).
[4]Fermi GBM Team, "GRB 240615A: Fermi GBM Final Real-time Localization," *GRB Coordinates Network*, vol. 36671, p. 1, Jun. 2024, Accessed: Jul. 31, 2024. [Online]. Available: https://ui.adsabs.harvard.edu/abs/2024GCN.36671....1F/abstract
[5]G. Pugliese *et al.*, "The red optical afterglow of GRB 030725," *Astronomy and Astrophysics*, vol. 439, no. 2, pp. 527–532, Jul. 2005, doi: https://doi.org/10.1051/0004-6361:20052948.
[6]K. Hensley, "Featured Image: How Close Is Too Close to a Kilonova?," *AAS Nova*, Feb. 05, 2024. https://aasnova.org/2024/02/05/featured-image-how-close-is-too-close-to-a-kilonova/ (accessed Aug. 01, 2024).
[7]M. Diaz, G. Gonzalez, and J. Pullin, *The Sounds of the Cosmos*. MIT Press, 2023.

Results

The CTMO pipeline generates three main products from the data files: reduced science images, the limiting magnitude and a catalog of all the magnitudes observed in the science images, and a magnitude-color diagram. Figure 5 and Figure 6 serve as a before-and-after comparison of the calibration process. The limiting magnitude provides a quantity for depth and the catalog of observed magnitudes is used for cross-matching with the Gaia survey. The limiting magnitude was measured to be m = 21.5, which can be converted to a limiting sensitivity distance of the telescope using the distance modulus.

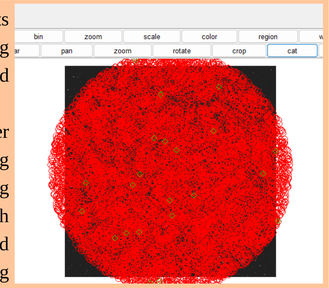


Fig. 6: Catalog cross-matching in ds9 using Tycho-2 (optical) and 2MASS point sources (IR).

An example of the catalog cross-matching is demonstrated in Figure 7. With relative photometry, a magnitude-color diagram is produced, alongside with quantities that measure the conditions of sky on the night of observation. These quantities are the extinction, which represents the effects of the gas and dust between the observer and object on observation, and the zeropoint, which compensates for the difference in observed magnitude. As a result of this analysis, it could be concluded that GRB240615 was not observed. However, GRB240615 was observed by other ground-based observatories in the J-band filter [2].

Discussion

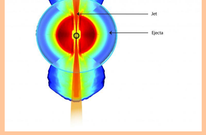


Fig. 7: An illustration of how the observing angle may influence the received signal [6].

There are several factors that affect the visibility of a GRB and the strength of the signal observed. GRB visibility can depend on the nature of the GRB and the weather and climate conditions of the observing location. GRB240615a was measured to have a relatively high redshift (lower limit z > 4.5), meaning the object is a distance that exceeds the viewing limitations of CTMO [2]. Further considerations regarding the obstruction of visibility include the viewing angle and the region which the GRB was localized, as conveyed in Figure 8. GRB240615a was localized to be near the galactic plane, a region of the Milky Way's disk where stars, dust, etc. is highly concentrated.

Regarding the weather, the night of observation was reported to be humid with a bright moon in the sky, increasing the difficulty to perform deep sky imaging.

Conclusion

We can confirm that GRB240615a was not observed by CTMO in the optical spectrum on 2024 June 16. It is suspected that the inability to observe the optical afterglow of GRB240615a was due to diminished viewability. Reasoning includes i) the characterization of the transient and ii) environmental interferences. The redshift and the region of localization of GRB240615a insinuates it to be of faint natures, heightening the difficulty of observation. Moreover, the weather and sky conditions on the night of observation were not ideal for deep sky imaging. In the future, CTMO aims to continue its GRB detection by following up NASA GCN alerts and to improve and expand upon the pipeline for data analysis.

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